

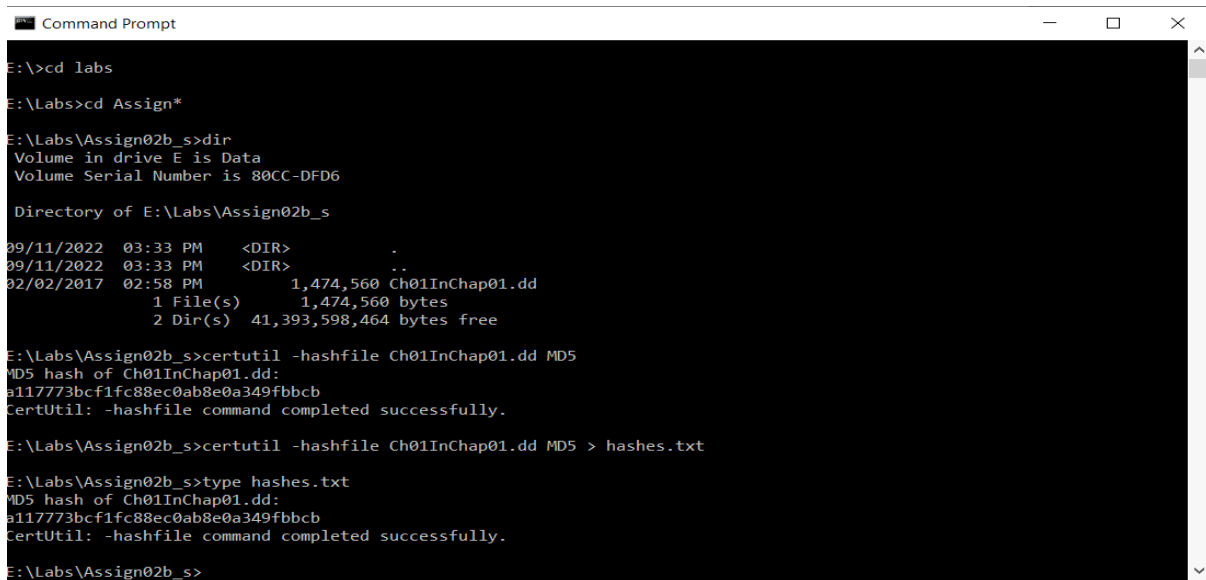
This assignment is content as the teacher said in lecture from taking screenshot and homework

❖ Screenshots with explanation

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❖ questions

- 1- One-page answers to question in assignment



```
Command Prompt
E:\>cd labs
E:\Labs>cd Assign*
E:\Labs\Assign02b_s>dir
Volume in drive E is Data
Volume Serial Number is 80CC-DFD6

Directory of E:\Labs\Assign02b_s

09/11/2022  03:33 PM    <DIR>          .
09/11/2022  03:33 PM    <DIR>          ..
02/02/2017  02:58 PM             1,474,560 Ch01InChap01.dd
               1 File(s)              1,474,560 bytes
               2 Dir(s)  41,393,598,464 bytes free

E:\Labs\Assign02b_s>certutil -hashfile Ch01InChap01.dd MD5
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.

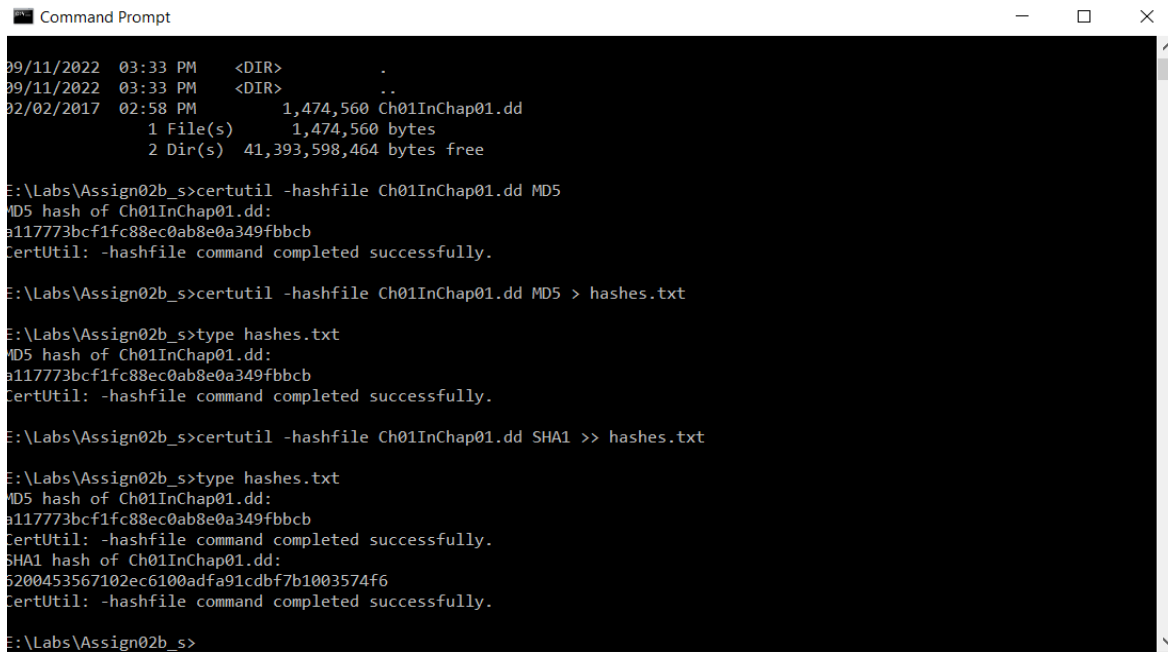
E:\Labs\Assign02b_s>certutil -hashfile Ch01InChap01.dd MD5 > hashes.txt

E:\Labs\Assign02b_s>type hashes.txt
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.

E:\Labs\Assign02b_s>
```

Figure 1: a screenshot of the command prompt window for calculating MD5 hash functions

This screenshot shows navigating to my E:\Labs\Assign02b_s folder by using the cd command and then calling certutil command for calculating the MD5 hash function the of the disk image, which I put result value inside of text file named hashes to use for matching with MD5 of forensic image when I calculate MD5 hash value to use as proof that bit by bit image matches of the disk image. In addition, I called the hashes.txt by type command to ensure the output of the MD5 hash value saved



```
09/11/2022 03:33 PM <DIR> .
09/11/2022 03:33 PM <DIR> ..
02/02/2017 02:58 PM          1,474,560 Ch01InChap01.dd
          1 File(s)          1,474,560 bytes
          2 Dir(s)  41,393,598,464 bytes free

E:\Labs\Assign02b_s>certutil -hashfile Ch01InChap01.dd MD5
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.

E:\Labs\Assign02b_s>certutil -hashfile Ch01InChap01.dd MD5 > hashes.txt

E:\Labs\Assign02b_s>type hashes.txt
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.

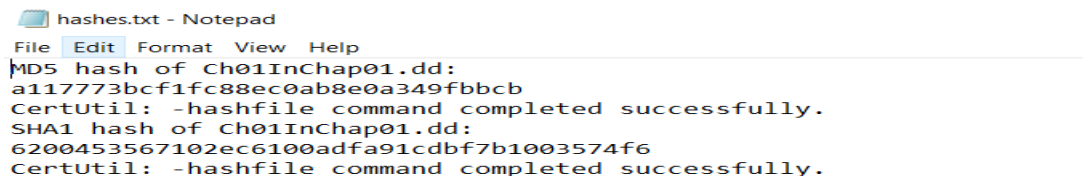
E:\Labs\Assign02b_s>certutil -hashfile Ch01InChap01.dd SHA1 >> hashes.txt

E:\Labs\Assign02b_s>type hashes.txt
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.
SHA1 hash of Ch01InChap01.dd:
6200453567102ec6100adfa91cdbf7b1003574f6
CertUtil: -hashfile command completed successfully.

E:\Labs\Assign02b_s>
```

Figure 2: a screenshot of the command prompt window for calculating the SHA1 hash function

This screenshot is continuing working on hash value in the middle of it calling certutil command for calculating the SHA1 hash function of the disk image, which I put the output in the same file of the previous function to use SHA1 for matching with SHA1 of forensic image when I calculate SHA1 hash value to use as proof that bit by bit image matches of the disk image. In addition, I called the hashes.txt by type command to ensure the output of SHA1 hash value saved



```
hashes.txt - Notepad
File Edit Format View Help
MD5 hash of Ch01InChap01.dd:
a117773bcf1fc88ec0ab8e0a349fbbcb
CertUtil: -hashfile command completed successfully.
SHA1 hash of Ch01InChap01.dd:
6200453567102ec6100adfa91cdbf7b1003574f6
CertUtil: -hashfile command completed successfully.
```

Figure 3: a screenshot of the file with the content of two hash functions of Ch01InChap01.dd

This screenshot shows the file that contains the output of two hash functions (MD5, SHA1) of the disk image completed successfully. I store the output of the calculation inside the file by using the command prompt window.

```
Cloning Log 2022-09-13, 10_30_10.txt - Notepad
File Edit Format View Help
09/13/2022, 10:30:10.8
E:\Labs\Assign02b_s\Ch01InChap01.dd --> E:\Labs\Assign02b_s\Ch01InChap01_WinHex.dd
Beginning of file --> Beginning of file
2,880 Sectors

09/13/2022, 10:30:10.8
2,880 sector(s) successfully copied.
```

Figure 4: a screenshot of a forensic copy of the disk image.

This screenshot shows successfully making a forensic copy of a disk image by using the WinHex program

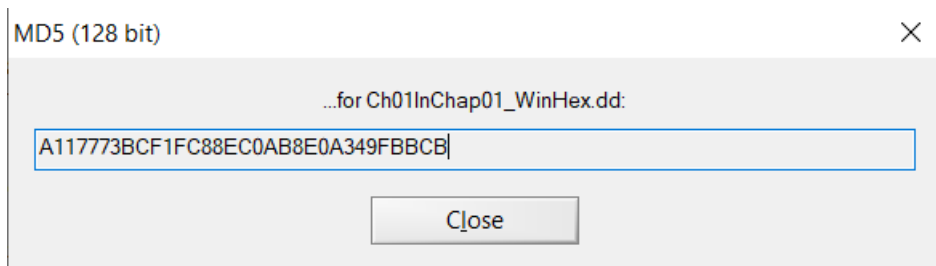


Figure 5: A screenshot of the MD5 hash function of the forensic image by WinHex tool

This screenshot shows the output of the calculation of the MD5 hash function of forensic copy by using the WinHex program to use as evidence that this result matches the disk image.

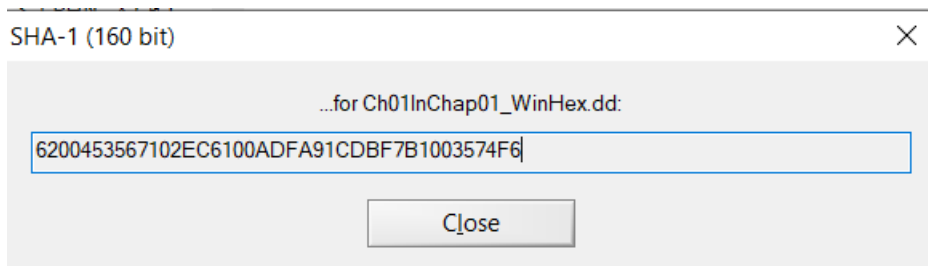


Figure 6: a screenshot of SHA-1 hashes function of the forensic image by WinHex tool

This screenshot shows the output of the calculation of SHA-1 hash function of forensic copy by using the WinHex program to use as evidence that this result matches the original image.

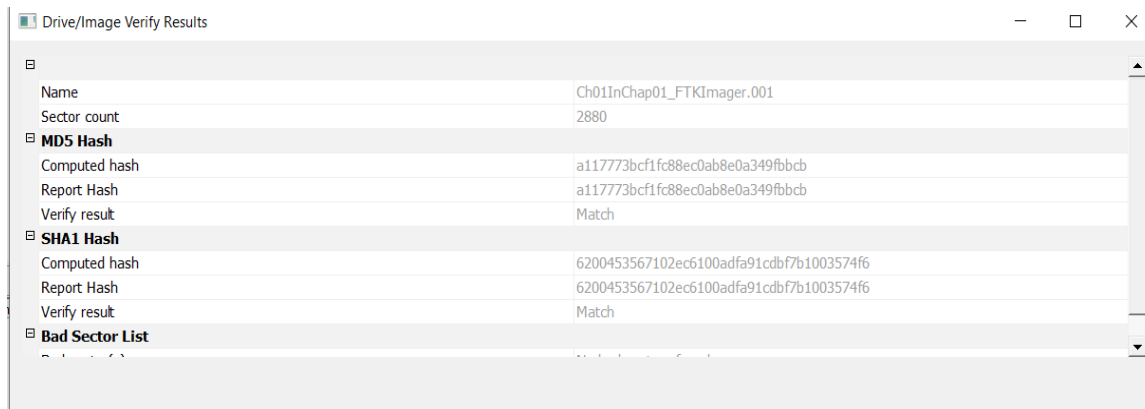


Figure 7: A screenshot of MD5 and SHA1 hash functions of the forensic image by FTK imager program

This screenshot shows the outputs of calculation of two hash functions of forensic copy by using FTK imager program to use as evidence that numbers that tell us that it is a bit-by-bit copy of disk image

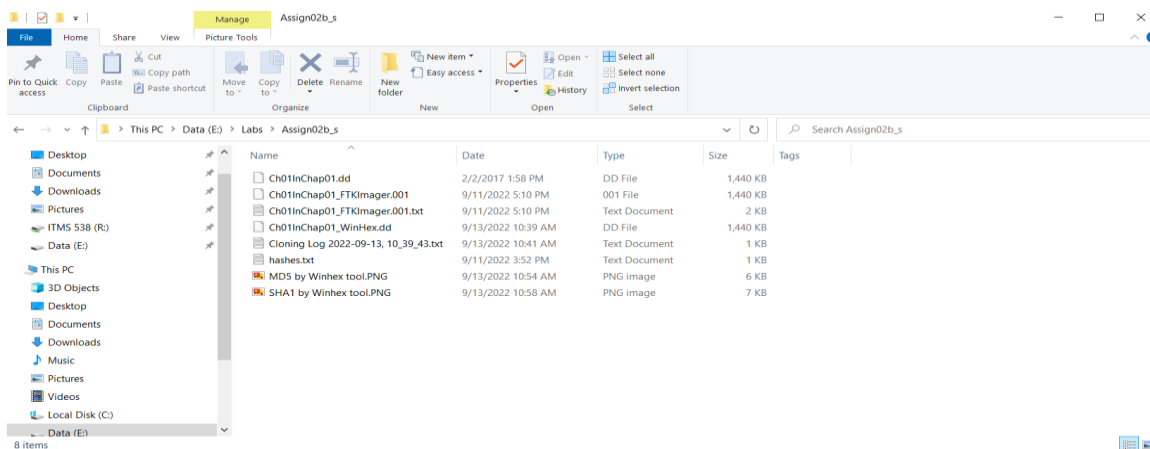


Figure 8: a screenshot of Assign02b_s folder under (E) drive

This screenshot shows various files, which are disk image by dd extension, forensic image by FTK imager with its format, an exact copy of original sources by WinHex program, hashes file content of MD5 and SHA1 of the original image which made by command prompt window, Cloning log shows the forensic copy successful by WinHex tool, two pictures of hash functions (MD5-SHA1) by WinHex program.

- What you accomplished in this lab

1-In this lab; a student created a Forensic image of a disk image, which means I created bit by bit image using tools called image files which assist in making an exact copy by various programs such as FTK imager, WinHex.

2- I calculated two Hash functions by command Prompt of the disk image and forensic images by WinHex and FTKImager programs and verified the outputs to check that the exact copy matches the original image

3- opening the file picture, which I knew from the extension by changing the signature of the file.

- A short description of the SHA1 and MD5 hash values and why they were used to verify the forensic images

SHA1 and MD5 are hash functions that use for calculating to provide a stream of bits that use the output of the calculation as prove that the original image matches the forensic image.

(Input) → hash functions (MD5) = (output) (original image)

(Input) → hash functions (SHA1) = (output)

Match

(Input) → hash functions (MD5) = (output) (forensic image)

(Input) → hash functions (SHA1) = (output)

- How the MD5 and SHA1 hash functions differ (at a high level) and why you think these were picked

The two hash functions are different from each other's because when I calculate the hash value the output of MD5 different from output of SHA1 and I match the output of MD5 of original storage medium with MD5 of forensic image and SHA1 hash value of original storage medium with SHA1 forensic image. The investigators must not work on the digital evidence that found, so they must make forensic copy of original storage medium to retrieve the data and working on it to prove as evidence in court and to ensure that forensic copy match the original source the examiners should provide that by calculate two functions of hash values which helps them to use as evidence that forensic image which makes by tools exactly match original storage medium. In addition, these hash functions are designed so that even one bit changed in the file that will provide a very various match value.

- An explanation as to why we didn't just use a single hash function for verification

Each original storage medium and Forensic image has hash functions (SHA1, MD5) that work as their identities. Therefore, two hash functions are required, which cannot do one without the other that must compare both functions of the original storage medium and the exact copy to ensure by calculating that stream of bits inform us that it is a bit-by-bit copy